



Surya Narayana Murthy Pilli

surya.p2805@gmail.com | + 1 (615)-960-4713

www.linkedin.com/in/surya-n-m-p-836151184

Professional Summary:

- Full Stack Java Developer with 7+ years of experience designing, developing, and delivering enterprise-scale web applications across healthcare, supply chain, and finance domains.
 - Expert in **Java, Spring Boot, Spring Cloud, Hibernate, and Spring Security**, building resilient, secure, and scalable microservices with service discovery, centralized configuration, circuit breakers, and OAuth2/JWT-based authentication.
 - Skilled in **React.js and Redux**, creating responsive, modular, and user-centric applications with reusable components, hooks, and efficient state management.
 - Experienced in **Angular**, developing dynamic, component-driven applications using reactive forms, RxJS, dependency injection, and optimized performance patterns.
 - Proficient in **TypeScript, HTML5, CSS3, JavaScript, and modern frontend tooling**, delivering maintainable and high-performance web applications.
 - Hands-on experience with **AWS (EC2, S3, RDS, Lambda, Elastic Beanstalk, CloudFormation, Terraform, Kubernetes)** for cloud-native deployments, automation, and Infrastructure as Code (IaC).
 - Experienced with **Azure (App Services, Functions, SQL, Blob Storage, ARM templates, Logic Apps)**, implementing scalable, resilient, and secure cloud solutions.
 - Strong expertise in **RDBMS (Oracle, MySQL, PostgreSQL)** and **NoSQL (MongoDB, Cassandra, DynamoDB)**, with ORM skills in Hibernate and JPA to optimize database performance and scalability.
 - Skilled in **RESTful and SOAP web services**, implementing **Swagger/OpenAPI** for integration, testing, and API documentation.
 - Experienced in **unit and UI testing frameworks** including JUnit, Mockito, Karma, Jasmine, and Protractor, ensuring robust, production-ready applications.
 - Proficient in **CI/CD and DevOps pipelines** using Git, GitHub, Maven, Gradle, Jenkins, SonarQube, Docker, and container orchestration with Kubernetes.
 - Experienced in **secure application development**, implementing RBAC, TLS/HTTPS, JWT authentication, and adhering to **OWASP Top 10, HIPAA, and PCI-DSS** compliance standards.
 - Skilled in **messaging and event-driven architectures** using Kafka, RabbitMQ, and ActiveMQ, enabling real-time and fault-tolerant systems.
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Education:

Masters in VLSI Technology, CMR Institute of Technology, Hyderabad, India

Bachelors in Electronics and Communication Engineering., Jawaharlal Nehru Technological University, Hyderabad, India

Technical Skills:

Languages	Java, JavaScript, Python, C, C++
Backend/ Frameworks	Spring, Spring Boot, Spring MVC, Spring Security, Spring Cloud, Spring Data, Hibernate, JPA, JDBC, Servlets, JSP, WebFlux, Node.js, Express.js
Frontend / Web	Angular (2+), ReactJS, Redux, Vue.js, HTML5, CSS3, Bootstrap, Sass, AJAX, JSON, XML
Databases	Oracle, SQL Server, MySQL, PostgreSQL, MongoDB, Cassandra, DynamoDB
Cloud & DevOps:	AWS (EC2, S3, RDS, Lambda, Elastic Beanstalk, CloudFormation, Terraform), Azure (App Services, Functions, SQL, Blob Storage, ARM Templates), Jenkins, GitLab, AWS CodePipeline, Maven, Gradle, Docker, Kubernetes

Testing & Tools	JUnit, Mockito, Karma, Jasmine, Protractor, Cucumber, JProfiler, Log4j, SonarQube, Git, GitHub, CVS, SVN, GitHub Copilot
Cloud	AWS (EC2, S3, RDS, Lambda, Elastic Beanstalk, CloudFormation, Terraform), Azure (App Services, Functions, SQL, Blob Storage, ARM Templates)
Additional Expertise	SOAP Web Services, RESTful Services, Microservices Architecture, Multithreading, MVC Design Pattern, Reactive Programming, Agile/Scrum

Professional Experience:

Client: Natera, Pleasanton, CA

March 2025 - Present

Role: Senior Software Engineer

Responsibilities:

- Designed and developed **Angular Single Page Applications (SPAs)** using modern Angular (components, modules, Angular CLI), leveraging **RxJS** for reactive data flows and **NgRx** for predictable state management to ensure scalable, maintainable front-end architecture.
- Improved frontend performance with **lazy-loaded feature modules, route-level code splitting, ChangeDetectionStrategy.OnPush, trackBy usage**, and efficient observable piping to reduce render overhead and improve perceived responsiveness.
- Developed and maintained Java **Swing** based desktop applications, designing dynamic and interactive **GUIs** using layout managers, event listeners, and custom Swing components to deliver an intuitive, responsive user interface for internal business users.
- Refactored legacy **Swing** modules into modular MVC/MVP architecture, improving maintainability, reducing code coupling, and integrating backend data access through JDBC and RESTful APIs to synchronize real-time data between desktop and web platforms.
- Built **reusable, accessible UI component libraries** with Angular Material and custom component libraries (standalone components + reactive forms), enforcing design systems and cross-application consistency.
- Integrated **REST and GraphQL APIs** with Angular HttpClient, interceptors, and Apollo Angular, optimizing client-server communication with caching, pagination, and error handling strategies.
- Implemented **server-side rendering (SSR)** and SEO improvements using Angular Universal to increase crawlability and initial paint performance for customer-facing portals.
- Wrote **unit and integration tests** using Jest (or Jasmine/Karma where required) and end-to-end tests with Cypress, driving high test coverage and reliable release quality.
- Engineered **scalable and resilient Spring Boot microservices** in Java, implementing fault-tolerant patterns and domain-driven service design for distributed systems.
- Applied **Spring WebFlux** for reactive services and **Spring MVC** for synchronous endpoints, selecting the appropriate programming model based on service SLA and throughput characteristics.
- Leveraged **Spring Data JPA and Hibernate** for robust data access to Oracle (and other RDBMS), using named queries, stored procedures, and batch operations to significantly improve large-batch performance.
- Implemented **Spring Security** with OAuth2 / JWT for token-based and role-based authentication and authorization across microservices.
- Used **Spring Cloud components** (Eureka, Config Server, API Gateway, Feign clients) for service discovery, centralized configuration, and simplified inter-service communication.
- Designed **circuit breaker and fallback logic** using Resilience4j, ensuring graceful degradation and high availability of critical features.
- Designed **REST APIs and OpenAI/Swagger specifications** to standardize interfaces and improve cross-team collaboration.
- Containerized **Spring Boot services with Docker**, deployed and orchestrated on **Kubernetes** (Helm, liveness/readiness probes), enabling auto-scaling and resilient production deployments.
- Automated **AWS infrastructure provisioning** with Terraform, managing EC2, RDS, S3, CloudWatch, and serverless integrations for production-grade cloud environments.
- Configured **AWS RDS** with backup, read replicas, and disaster recovery patterns to ensure high availability for relational data.
- Leveraged **AWS Lambda** for event-driven, serverless processing to reduce infrastructure costs and handle asynchronous workflows (S3 triggers, SNS/SQS integrations).
- Built **CI/CD pipelines in Jenkins** (Git branching, Maven builds, SonarQube quality gates), enabling reliable automated builds, tests, and deployments.

Client: Electrum, LA, CA.

April 2022 – Jan 2025

Role: Senior Full Stack Developer

Responsibilities:

- Developed dynamic and responsive **UI modules using Angular**, leveraging a component-based architecture to ensure reusability and maintainability across multiple financial applications.
- Implemented Angular services and **RxJS observables** to handle asynchronous data and manage reactive state updates, improving real-time user interactions.
- Built **RESTful APIs using Spring Boot**, enabling seamless communication between front-end components and backend services while ensuring modular design and scalability.
- Implemented **MVC design pattern using Spring MVC**, separating model, view, and controller layers to enforce clean architecture and maintainable backend code.
- Designed **MongoDB schemas, collections, and indexes**, optimizing queries for high-volume transactions and ensuring scalable data storage solutions.
- Configured **Spring Security with JWT based authentication** and role-based access control, safeguarding both front-end and back-end layers from unauthorized access.
- Developed **batch processing and scheduled jobs using Spring Batch and Quartz Scheduler**, streamlining financial transaction processing and reporting.
- Integrated **asynchronous messaging using Azure Service Bus**, ensuring reliable event driven communication for high-volume financial operations.
- Built **CI/CD pipelines using GitHub Actions**, automating Gradle builds, testing, and deployments to Azure App Services for faster and error-free releases.
- Automated **infrastructure provisioning using CloudFormation templates**, ensuring repeatable and consistent deployment of Azure resources.
- Secured sensitive application credentials and encryption keys using **Azure Key Vault**, ensuring centralized secrets management and compliance with enterprise security standards.
- Conducted **unit testing for Angular components using Karma, Jasmine** and backend services using JUnit and Mockito, following TDD principles to ensure code quality.
- Performed **integration and end to end testing**, validating REST endpoints and Angular services to guarantee robust system functionality.
- Optimized backend performance with **multithreading, connection pooling, and efficient Hibernate ORM mappings**, reducing response times under high concurrency.
- Developed **reusable UI components and modular MVC controllers**, significantly reducing development effort for new features and maintaining consistency across applications.
- Implemented **robust logging and monitoring using Log4j2 and Azure Monitor**, enabling proactive detection and troubleshooting of performance bottlenecks and system issues.

Client: Kapton, Hyderabad, India.

October 2021 - March 2022

Role: Software Developer

Responsibilities:

- Developed **highly interactive React.js front-end modules** using functional components, hooks, context API, and Redux to manage complex application state efficiently.
- Integrated advanced **data visualization in React** using charting libraries (**Chart.js, Recharts**), enabling providers to analyze patient statistics and healthcare metrics in real time.
- Implemented **dynamic form handling and validation logic**, ensuring real-time user feedback and data integrity across multiple patient workflows.
- Built **modular and reusable UI components**, improving consistency and accelerating feature development across the platform.
- Developed **backend services with Spring Boot**, applying MVC principles to maintain clear separation between model, view, and controller layers.
- Implemented custom schedulers in **Spring Boot** to automate recurring healthcare processes, such as appointment reminders and billing notifications, reducing manual intervention.
- Integrated **SOAP web services** for secure and reliable communication with internal and external healthcare systems, handling XML data transformation and schema validation.
- Implemented **asynchronous message processing using JMS**, enabling high-volume transaction handling without blocking critical workflows.
- Designed **PostgreSQL schemas and optimized queries**, including stored procedures and triggers, to ensure efficient and reliable data operations.

Client: Accenture, Hyderabad, India.

July 2018– October 2021

Role: Associate Software Engineer

Responsibilities:

- Developed **dynamic JSP based forms** with advanced client-side validations using JavaScript, improving user interaction and reducing input errors.
- Built **modular presentation layer components** with rich UI elements to standardize look and feel across healthcare management workflows.
- Designed and implemented **Java modules encapsulating business rules and workflow logic**, ensuring maintainable and reusable backend code.
- Published and consumed **SOAP web services with WSDL**, performing XML message validation and implementing secure communication between enterprise systems.
- Implemented **Spring AOP for cross-cutting concerns** including logging, transaction management, and exception handling, enhancing maintainability.
- Developed **Dispatcher Servlets, Controllers, and View Resolvers** for MVC based application flow, improving request routing and response handling.
- Designed and manipulated **JSON and XML payloads** for integration with Cassandra database, supporting distributed and highly available storage of healthcare records.
- Optimized **Cassandra data modeling**, creating tables with composite keys and clustering strategies to support high-volume healthcare transactions.
- Applied **Hibernate ORM and Spring DAO patterns** to manage persistence while integrating with Cassandra through custom repositories.
- Implemented **advanced TDD practices using JUnit**, building reliable unit and integration tests to enforce quality and reduce defects.
- Managed **source control with Git**, performing branching, merges, and collaborative workflows for distributed development teams.
- Applied **design patterns (Factory, Singleton, DAO) and modular coding practices** to improve code reusability and system scalability.